BCSE205L
Computer Architecture and Organization

Module 5 – Interfacing and Communication

Module:5	Interfacing and Communication	5 Hours
I/O fundamentals: handshaking, buffering, I/O Modules - I/O techniques: Programmed I/O, Interrupt-driven I/O, Direct Memory Access, Direct Cache Access - Interrupt structures: Vectored and Prioritized-interrupt overhead - Buses: Synchronous and asynchronous - Arbitration.		

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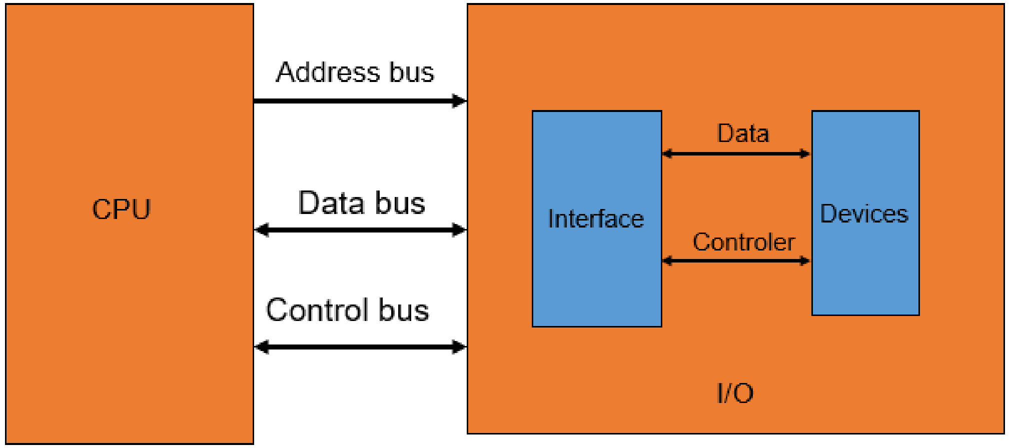
Introduction to I/O Module

- I/O is referred as Input-Output of a computer
- I/O devices are connected to CPU through I/O module called as Interfacing
- What for I/O?
  - I/O module provides an efficient mode of communication between CPU and outside of environment
- Why to use I/O module? Why can’t connect peripheral devices directly to the subsystem?
  - Diversity
  - Speed
  - Control

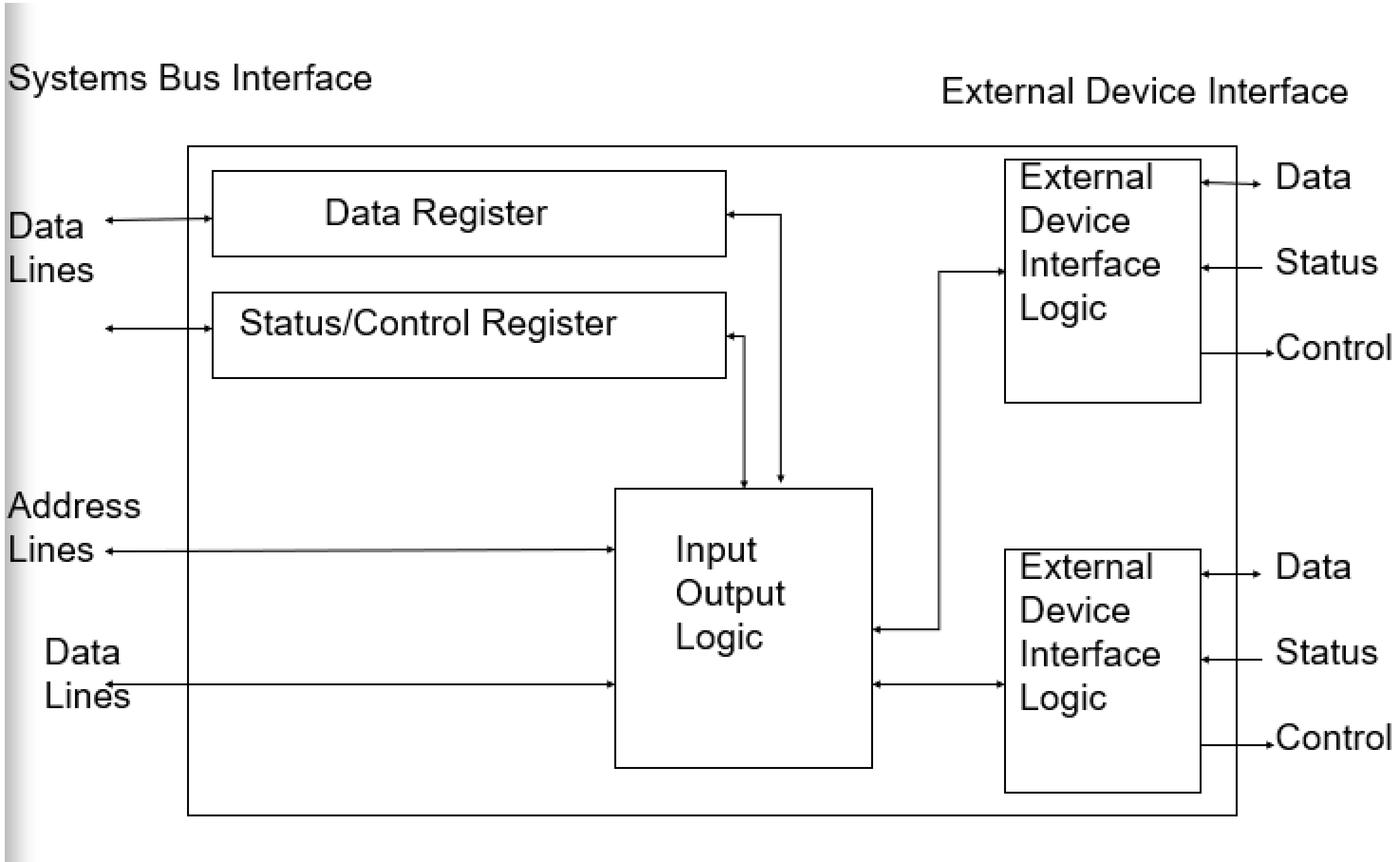
Functions of I/ O Module

- **Processor communication**
  - Data transfer between the processor and an I/O module
  - Accepting and decoding commands sent by the processor
  - Reporting of current status
- **Device communication** – It needs to be able to perform standard device communications, such as reporting of status.
- **Control and timing** – An I/O module needs to be capable of managing data flow between a computer's internal resources and any connected external devices.
- **Data buffering** – A crucial function that manages the speed discrepancy that exists between the speed of transfer of data between the processor and memory and peripheral devices.
- **Error detection** – Detecting errors, whether mechanical (such as a printer experiencing a paper jam) or data based, and reporting them to the processor

CPU & I/O Module Organization



- Many I/O devices
- Each device - unique address
- Processor- Command- address of IO device
- Processor, Main memory, IO - Share a common bus



Data Transfer

- **DATA TRANSFER**
  - The physical transfer of data (a digital bit stream) over a point-to-point or point-to-multipoint communication channel.
  - Example (channels): copper wires, optical fibres, wireless communication channels.
- **SYNCHRONOUS DATA TRANSFER**
  - The transfer of data between two devices on a network where they both carry out a predetermined set of interactions based on a common clock pulse.
- **ASYNCHRONOUS DATA TRANSFER**
  - The transfer of data between two devices on a network where they both carry out a predetermined set of interactions based on a private clock pulse.

Synchronous Data Transfers

- If the registers in the I/O interface share a common clock with CPU registers, then transfer between the two units is said to be **Synchronous**
- It usually occur when peripherals are located within the same computers as the CPU
- It shares a common clock

Asynchronous Data Transfers

- Asynchronous data transfer is a method of data transmission where data is sent in a non-continuous, non-synchronous manner. This means that data is sent at irregular intervals, without any specific timing or synchronization between the sending and receiving devices.
- Asynchronous Data Transfer between two independent units requires that control signals be transmitted between the communicating units so that the time can be indicated at which they send data.
- **Control signals**
    - **Strobe control:** A strobe pulse is supplied by one unit to indicate to the other unit when the transfer has to occur.
    - **Handshaking:** This method is commonly used to accompany each data item being transferred with a control signal that indicates data in the bus. The unit receiving the data item responds with another signal to acknowledge receipt of the data.

Asynchronous Data Transfer Methods

Methods to accomplish Asynchronous Data Transfer

i. Source Initiated Strobe for Data Transfer.

ii. Destination Initiated Strobe for Data Transfer.

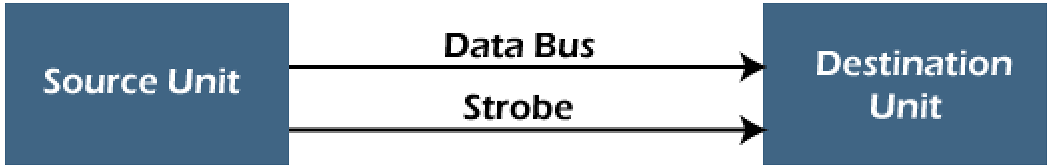
iii. Source Initiated Transfer Using Handshaking.

iv. Destination Initiated Transfer Using Handshaking.

Asynchronous Data Transfer Methods

i) Source Initiated Strobe for Data Transfer

- Source unit places data on the data bus and activates Strobe pulse.
- Data bus carries binary information from source to destination.
- Information on Data bus remains in Active state till the data is received by destination.
- Destination unit uses falling edge to transfer data.
- Source removes data from the bus and disables the Strobe pulse.



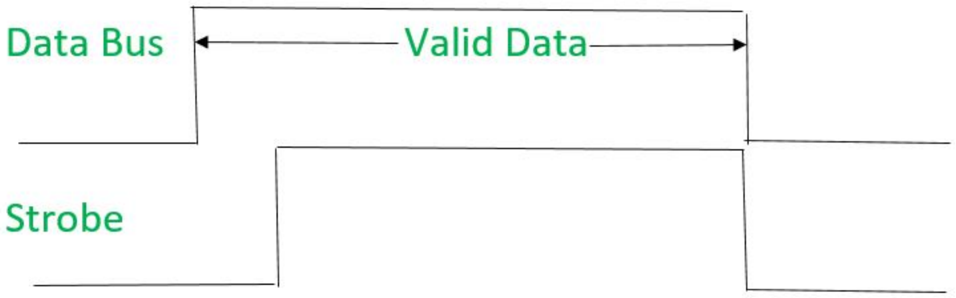
(a) Block Diagram

(i) First, source puts data on the data bus and ON the strobe signal

(ii) Destination on seeing the ON signal of strobe, read data from the data bus.

(iii) After reading data from the data bus by destination, strobe gets OFF.

Fig. shows that first data is put on the data bus and then strobe signal gets active.



Asynchronous Data Transfer Methods

ii) Destination Initiated Strobe for Data Transfer.

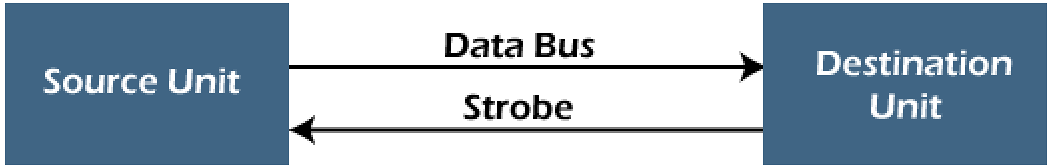
- Destination unit activates Strobe pulse.
- Source responds by placing the requested data on the bus.
- Data must be valid and available for small period of time, so destination can accept the information.
- Destination unit disables the Strobe pulse.
- Source removes the data from the bus.

(i) First, the destination ON the strobe signal to ensure the source to put the fresh data on the data bus.

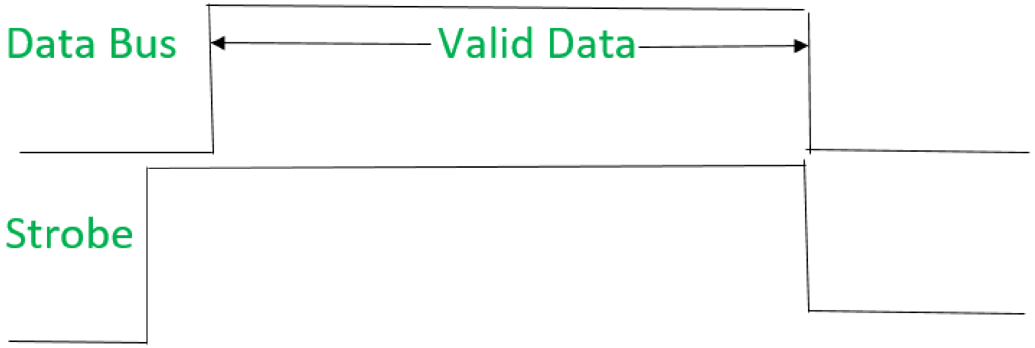
(ii) Source on seeing the ON signal puts fresh data on the data bus.

(iii) Destination reads the data from the data bus and strobe gets OFF signal.

Fig. shows that first strobe signal gets active then data is put on the data bus



(a) Block Diagram



Asynchronous Data Transfer Methods

- **Problems faced in Strobe based asynchronous**
  - **No Acknowledgement for Data Sent/Received.**
  - In Source initiated Strobe, it is assumed that destination has read the data from the data bus but there is no surety.
  - In Destination initiated Strobe, it is assumed that source has put the data on the data bus but there is no surety.
  - This problem is overcome by **Handshaking**

Asynchronous Data Transfer Methods

iii)Source Initiated Transfer Using Handshaking

- When source initiates the data transfer process.
- It consists of signals:
  - DATA VALID: if ON tells data on the data bus is valid otherwise invalid.
  - DATA ACCEPTED: if ON tells data is accepted otherwise not accepted.

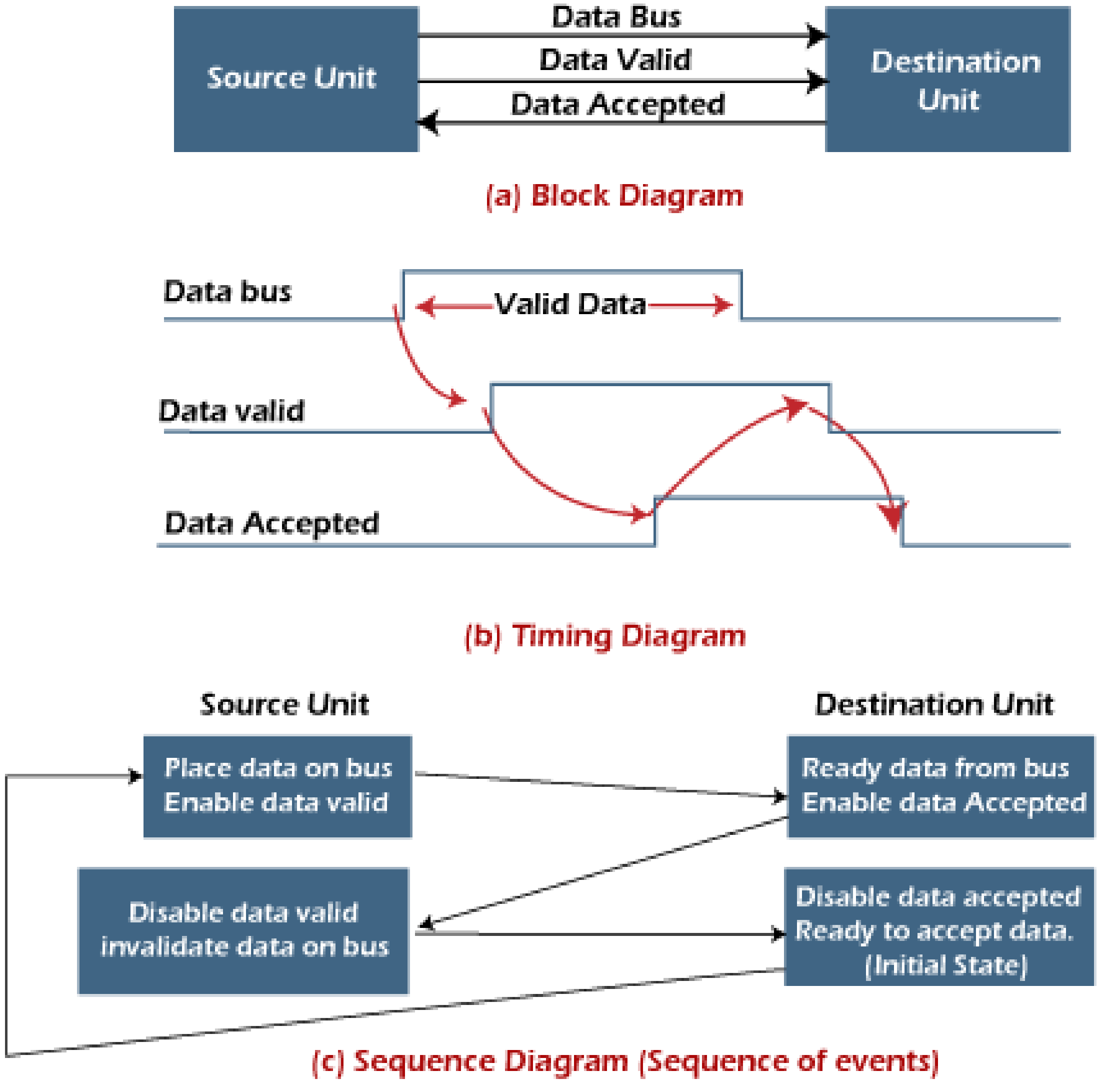
i) Source places data on the data bus and **enable Data valid** signal.

(ii) Destination accepts data from the data bus and **enable Data accepted** signal.

(iii) After this, **disable Data valid signal** - means data on data bus is invalid now.

(iv) **Disable Data accepted signal** and the process ends.

Now there is surety that destination has **read** the data from the data bus through data accepted signal.



Asynchronous Data Transfer Methods

iv) Destination Initiated Transfer Using Handshaking

- When destination initiates the process of data transfer.
- READY/REQUEST FOR **DATA**: if ON requests for putting data on the data bus.
- DATA VALID: if ON tells data is valid on the data bus otherwise invalid data.

  (i) When destination is ready to receive data, **Ready for Data signal gets activated**.

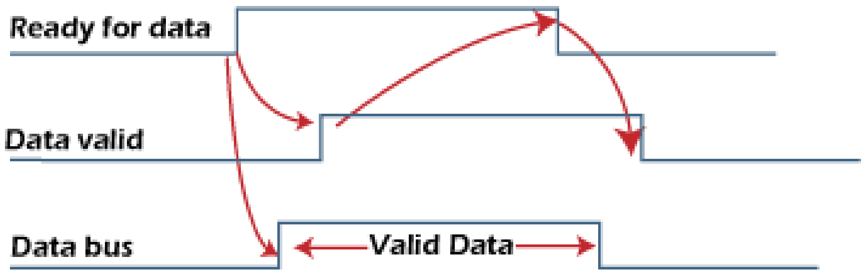
  (ii) Source in response **puts data on the data bus and enabled Data valid signal**.

  (iii) Destination then accepts data from the data bus and after accepting data, **disabled Request/Ready for Data signal.**

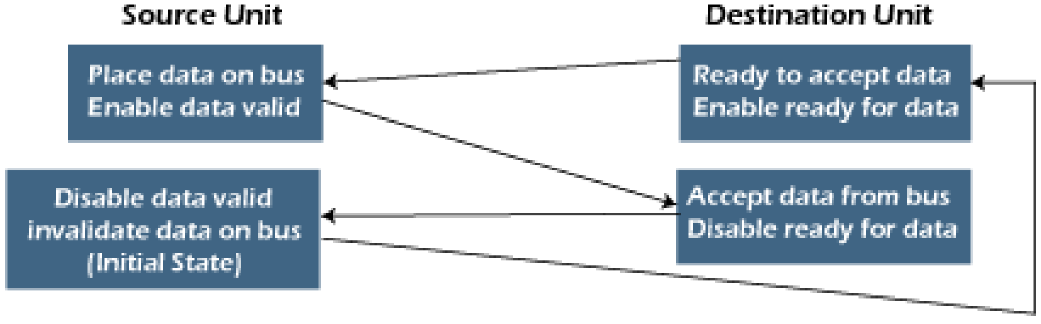
  (iv) At last, **Data valid signal gets disabled** means data on the data bus is no more valid data.

  Now there is surety that source has **put the data** on the data bus through data valid signal.

(a) Block Diagram



(b) Timing Diagram



(c) Sequence Diagram (Sequence of events)

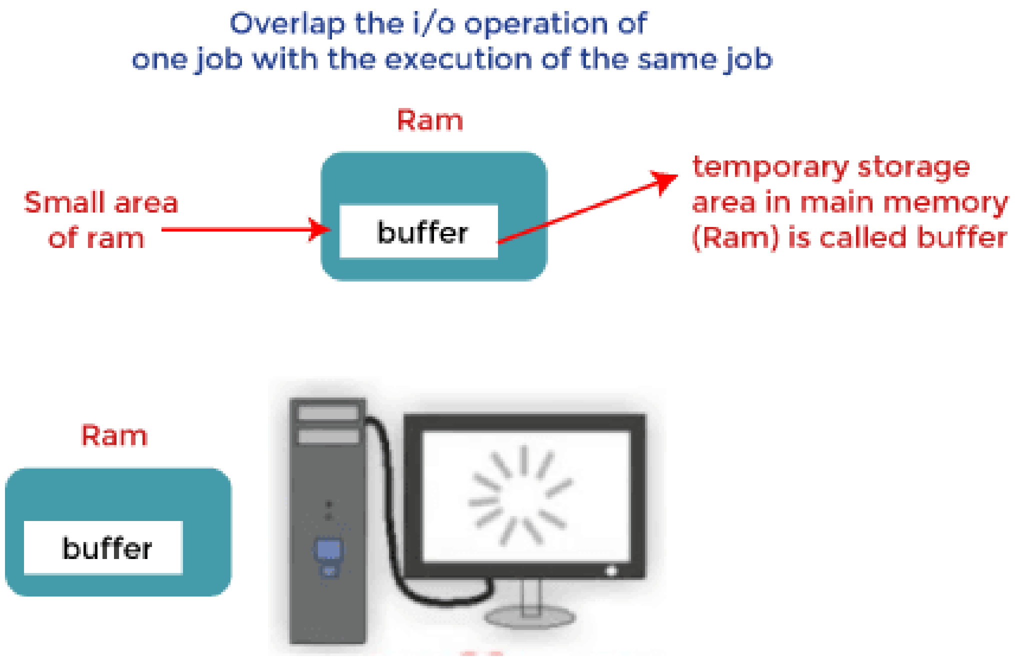
Asynchronous Data Transfer Methods

- Advantages:
  - Can be used to transfer any kind of data as it is flexible.
    (i.e. Text, Audio, video, etc)
  - Used in web applications like email, chatting.

Buffering

- Buffering is a technique that improves the efficiency and throughput of I/O operations.
- Plays a very important role in any OS during the execution of any process or task.
- Involves temporarily storing data while it's being transferred between two devices or processes.
- Buffering helps to
  - Synchronize two devices with different processing speeds
  - Help devices with different data transfer sizes adjust to each other
  - Support copy semantics
  - Smooth out peaks in I/O demand

How Buffering Works?

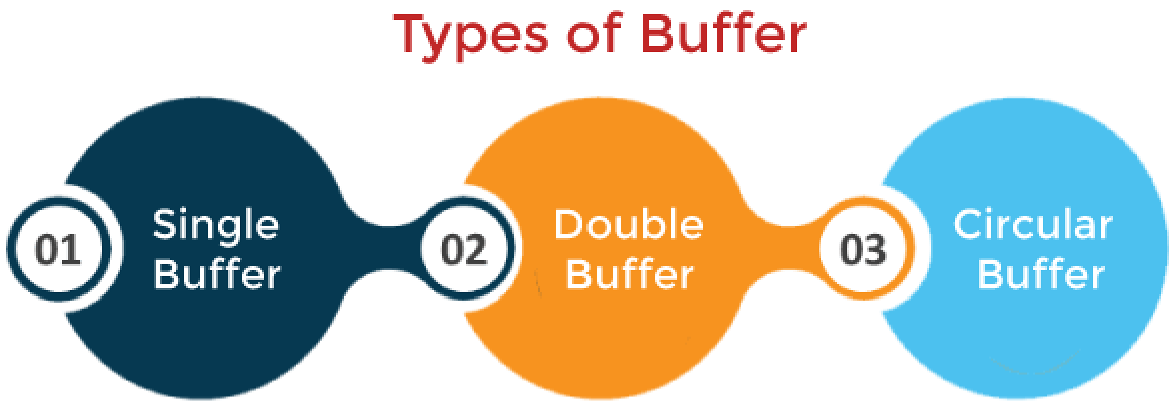


Buffering

Buffers - Implemented using a queue or FIFO algorithm in memory

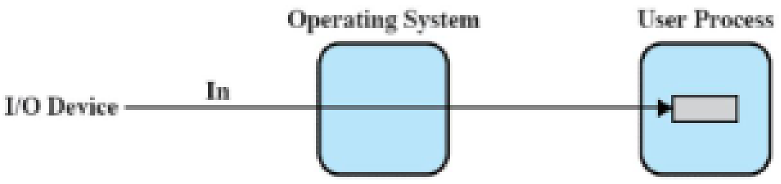
Types of Buffering

- Single buffering
- Double buffering
- Circular buffering

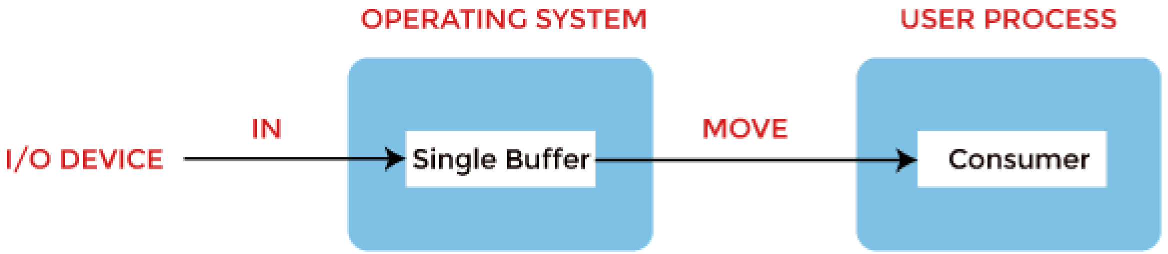


Types of Buffering

- **Single buffering**
  - In Single Buffering, only one buffer is used to transfer the data between two devices.
  - The producer produces one block of data into the buffer. After that, the consumer consumes the buffer.
  - Only when the buffer is empty, the processor again produces the data.



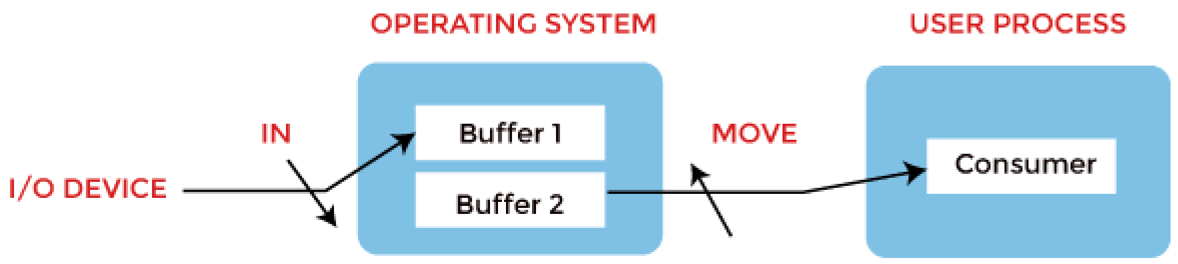
(a) No buffering



SINGLE I/O BUFFERING

Types of Buffering

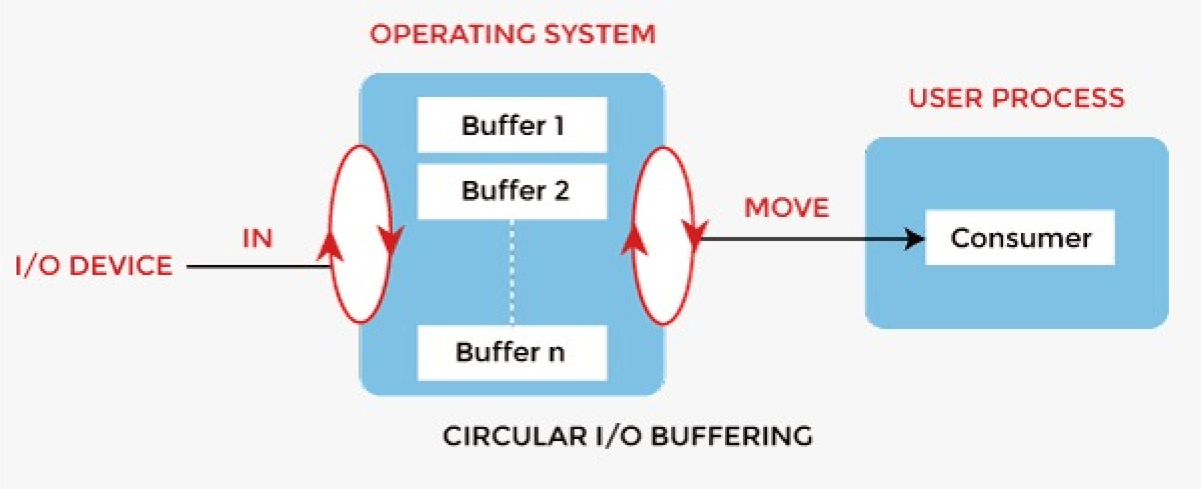
- **Double buffering**
  - Two schemes or two buffers are used in the place of one.
  - Producer produces one buffer while the consumer consumes another buffer simultaneously.
  - So, the producer not needs to wait for filling the buffer.
  - Double buffering is also known as buffer swapping.



DOUBLE I/O BUFFERING

Types of Buffering

- **Circular buffering**
  - More than two buffers are used, the buffers' collection is called a circular buffer.
  - Each buffer is being one unit in the circular buffer.
  - Increases data transfer rate than the double buffering.
  - Data do not directly pass from the producer to the consumer because the data would change due to overwriting of buffers before consumed.
  - The producer can only fill up to buffer x-1 while data in buffer x is waiting to be consumed



Advantages of Buffering

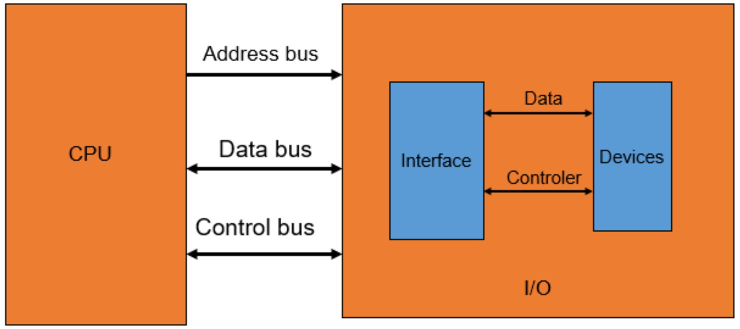
- Allows uniform disk access
- It simplifies system design
- The system places no data alignment restrictions on user processes doing I/O. By copying data from user buffers to system buffers and vice versa, the kernel eliminates the need for special alignment of user buffers, making user programs simpler and more portable.
- Reduce the amount of disk traffic – increases the overall system throughput and decreases response time.
- The buffer algorithms help ensure file system integrity.

Disadvantages of Buffering

- Costly
- Impractical to have the buffer be the exact size required to hold the number of elements. Thus, the buffer is slightly larger most of the time, with the rest of the space being wasted.
- Buffers have a fixed size at any point in time - When the buffer is full, it must be reallocated with a larger size, and its elements must be moved. Similarly, when the number of valid elements in the buffer is significantly smaller than its size, the buffer must be reallocated with a smaller size and elements be moved to avoid too much waste.
- Use of the buffer requires an extra data copy when reading and writing to and from user processes - when transmitting large amounts of data, the extra copy slows down performance.

I/O techniques

- I/O Interface: - Method used to transfer information between internal storage and external I/O devices
- CPU & Peripherals of a computer system – interfaced using special communication links - used to resolve the differences between CPU and peripheral
- Interface Units: - special hardware components between CPU and peripherals to supervise and synchronize all the input and output transfers



I/O interface

- Resolves the *differences* between the computer and peripheral devices

- Peripherals - Electromechanical Devices
- CPU or Memory - Electronic Device

- Data Transfer Rate
  - Peripherals - Usually slower
  - CPU or Memory - Usually faster than peripherals

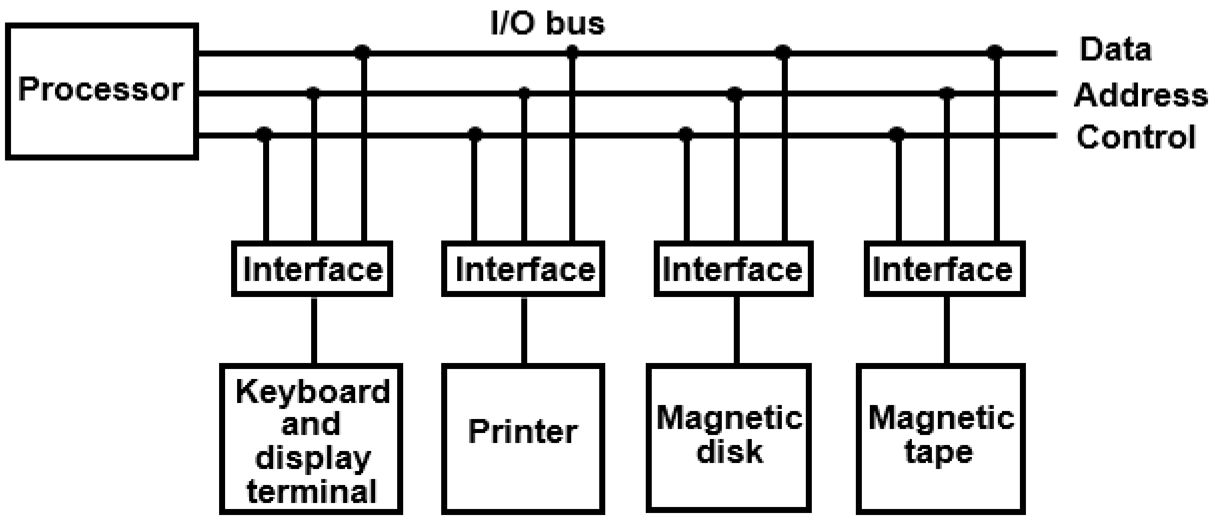
  Some kinds of Synchronization mechanism may be needed

- Unit of Information
  - Peripherals - Byte
  - CPU or Memory - Word

- Operating Modes
  - Peripherals - Autonomous, Asynchronous
  - CPU or Memory - Synchronous

I/O BUS AND INTERFACE MODULES

- Each peripheral has an interface module associated with it



Interface

- Decodes the device address (device code)
- Decodes the commands (operation)
- Provides signals for the peripheral controller
- Synchronizes the data flow and supervises

the transfer rate between peripheral and

CPU or Memory

Typical I/O instruction

Op. code	Device address	Function code
		(Command)

I/O techniques

- Data transfer to and from the peripherals may be done in any of the three possible ways
  - Programmed Input/Output
  - Interrupt driven Input/Output
  - Direct Memory Access

Technique	Interrupt Required	I/O module to/from Memory transfer
Programmed I/O	No	Through CPU
Interrupt driven I/O	Yes	Through CPU
Direct Memory Access	Yes	Direct transfer

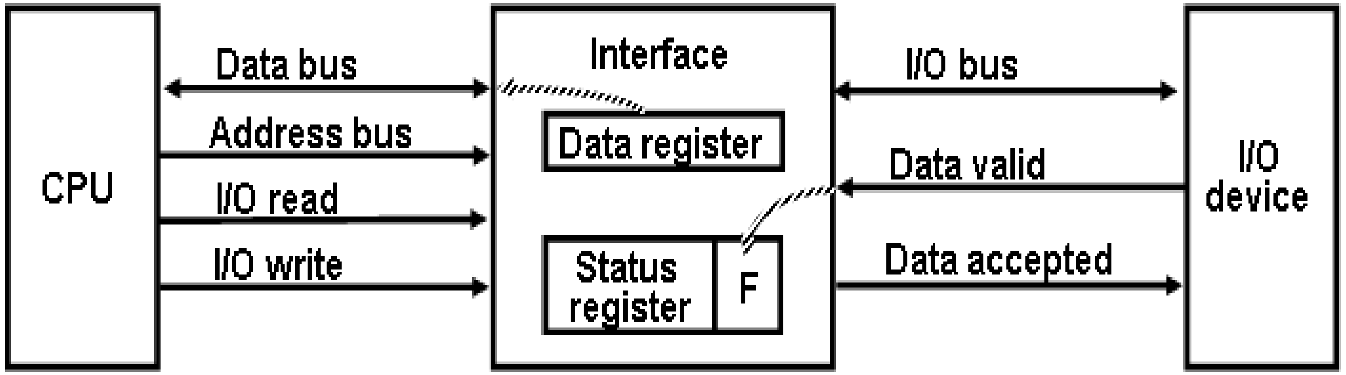
- Interrupt is a response by the processor to an event that needs attention from the software.

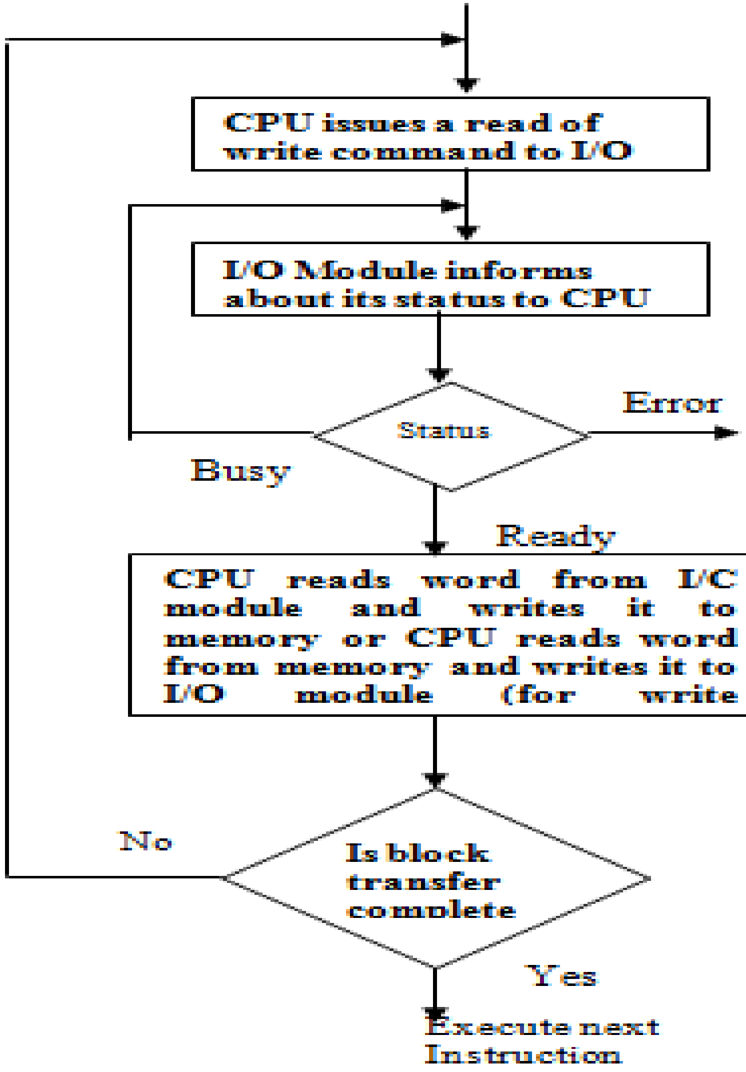
Programmed I/O

- It is due to the result of the I/O instructions that are written in the computer program.
- Each data item transfer is initiated by an instruction in the program
- Usually the transfer is from a CPU register and memory. Transferring data under this mode requires constant monitoring of the peripheral by CPU.
- Wastes CPU time
- **Example:** A transfer from I/O device to memory requires the execution of several instructions by the CPU involves the following steps.
  - Transfer of data from I/O device to the CPU registers
  - Transfer of data from CPU registers to memory
- **Disadv:** CPU monitors the I/O module and it is always busy - so that the CPU is not free to work on other things – can be avoided using INTERRUPTs

Programmed I/O - Process

1. CPU requests I/O operation
2. CPU must wait
3. I/O module performs operation
4. I/O module sets status bits
5. CPU checks status bits and then cleans up the operation





Programmed I/O - Process

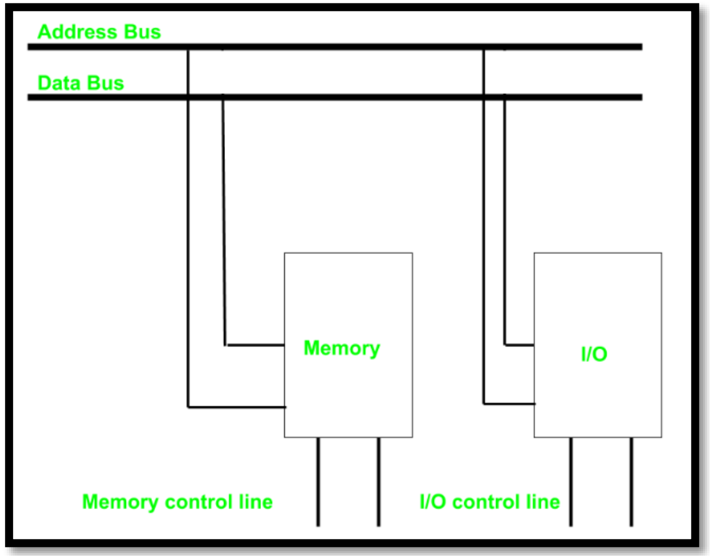
- I/O instruction in the program causes CPU to examine I/O module's control register's ready bit – if it is ready, then the CPU issues the I/O command by setting the proper bits in this register
- I/O module issues the command to the I/O device and watches over device
- When the I/O device has completed the task, the I/O module sets the proper bits in its control register
- CPU, watching this register, then cleans up the operation (on input, moves the input data from the I/O module's data register to the CPU register or memory location)

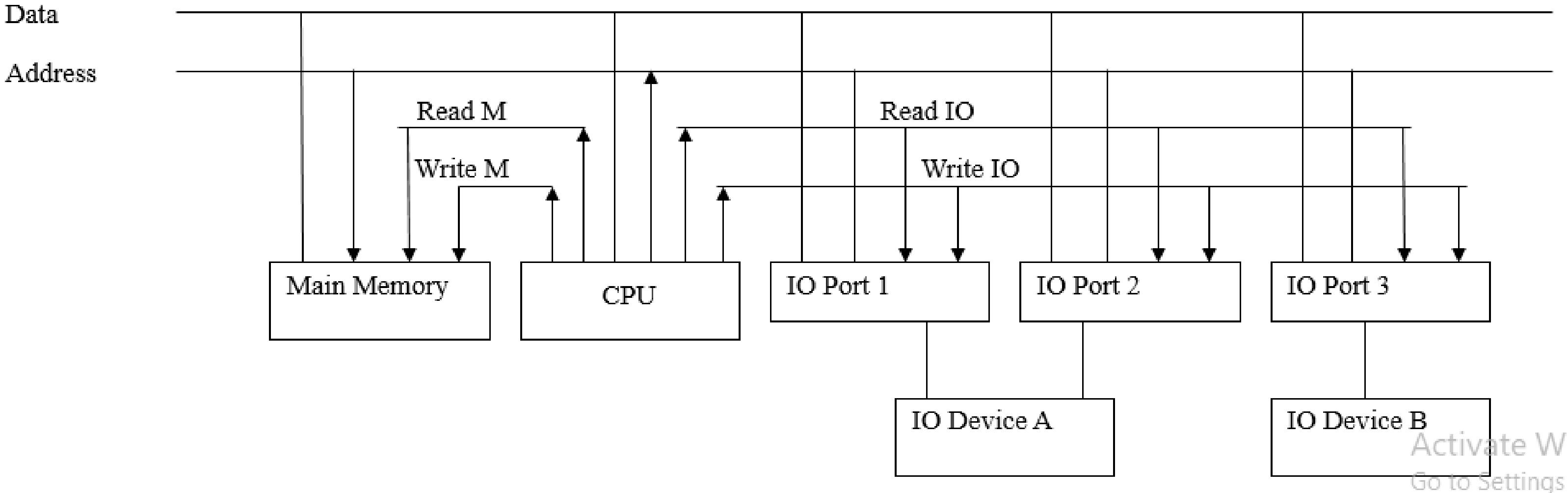
Programmed I/O – Addressing Methods

- As a CPU needs to communicate with the various memory and input-output devices (I/O) as we know data between the processor and these devices flow with the help of the system bus.
- There are three ways in which system bus can be allotted to them :
  - Separate set of address, control and data bus to I/O and memory.
  - Have common bus (data and address) for I/O and memory but separate control lines **(Isolated I/O)**
  - Have common bus (data, address, and control) for I/O and memory **(Memory Mapped I/O)**

Programmed I/O – Isolated I/O Addressing

- Common bus(data and address) for I/O and memory
- But separate read and write control lines for I/O
- So when CPU decode instruction, then if data is for I/O then it places the address on the address line and set I/O read or write control line on due to which data transfer occurs between CPU and I/O.
- Address space of memory and I/O is isolated – Isolated I/O.
- The address for I/O - called ports. Different read-write instruction for both I/O and memory.





I/O Mapped I/O

Programmed I/O – Isolated I/O Addressing

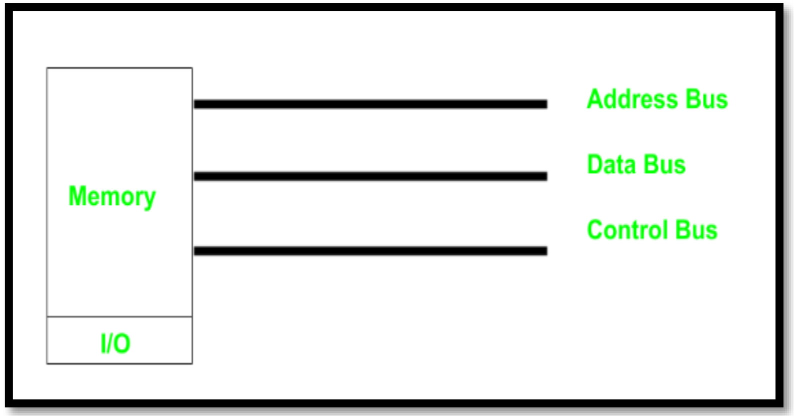
Advantages of Isolated I/O

- Large I/O Address Space
- Greater Flexibility
- Improved Reliability

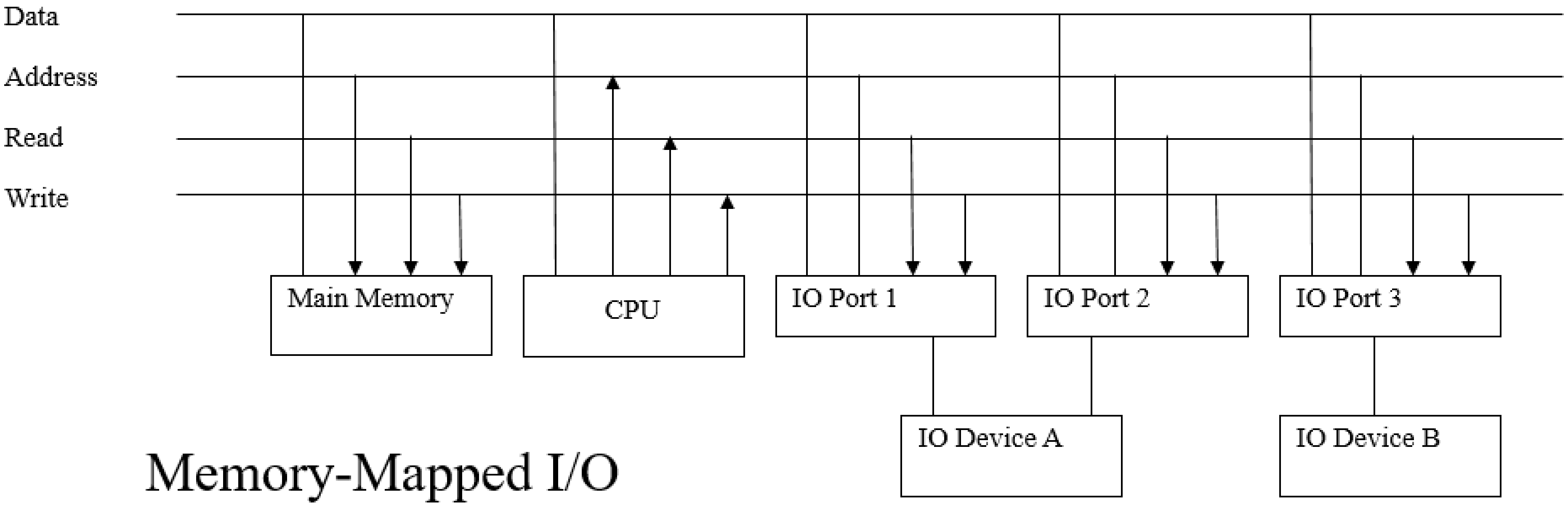
Disadvantages of Isolated I/O

- Slower I/O Operations
- More Complex Programming

Programmed I/O - Memory Mapped I/O Addressing



- Common bus (Address, Data, Control Bus) – Memory and I/O.
- I/O and memory - both have same address space
- **Disadv:** Addressing capability of memory become less because some part is occupied by the I/O.



Memory-Mapped I/O

Programmed I/O - Memory Mapped I/O Addressing

Advantages of Memory-Mapped I/O

- Faster I/O Operations
- Simplified Programming
- Efficient Use of Memory Space

Disadvantages of Memory-Mapped I/O

- Limited I/O Address Space
- Slower Response Time

Isolated I/O & Memory mapped I/O

Isolated I/O	Memory Mapped I/O
Memory and I/O have separate address space	Both have same address space
All address can be used by the memory	Due to addition of I/O addressable memory become less for memory
Separate instruction control read and write operation in I/O and Memory	Same instructions can control both I/O and Memory
In this I/O address are called ports.	Normal memory address are for both
More efficient due to separate buses	Lesser efficient
Larger in size due to more buses	Smaller in size
It is complex due to separate logic is used to control both.	Simpler logic is used as I/O is also treated as memory only.

Programmed I/O

Advantages

- Simple to implement
- Requires very little special software or hardware

Disadvantages

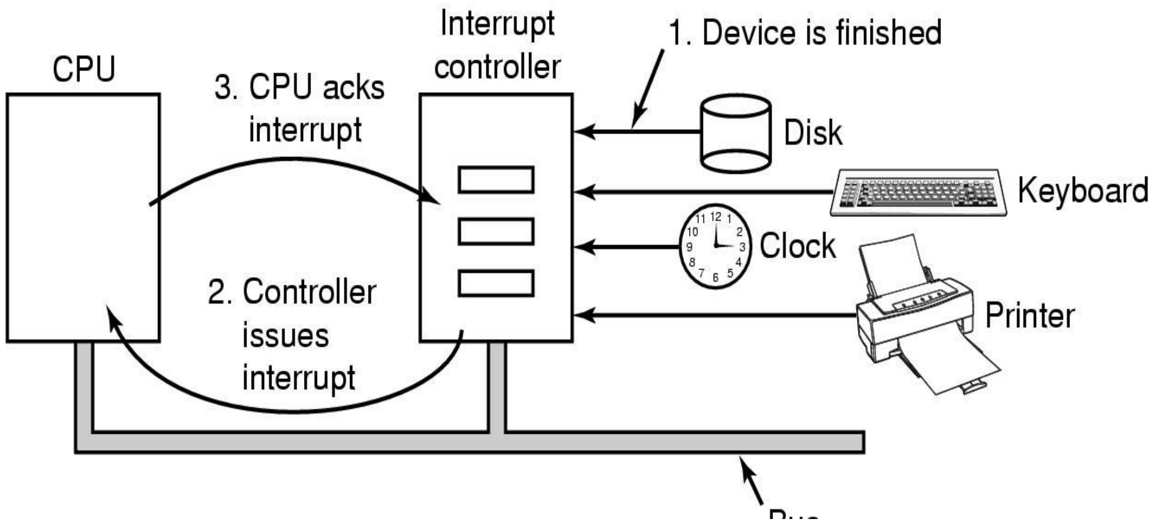
- Speed difference between CPU and the peripheral devices
- programmed I/O wastes an enormous amount of CPU cycles
- Very inefficient
- CPU slowed to the speed of the peripheral

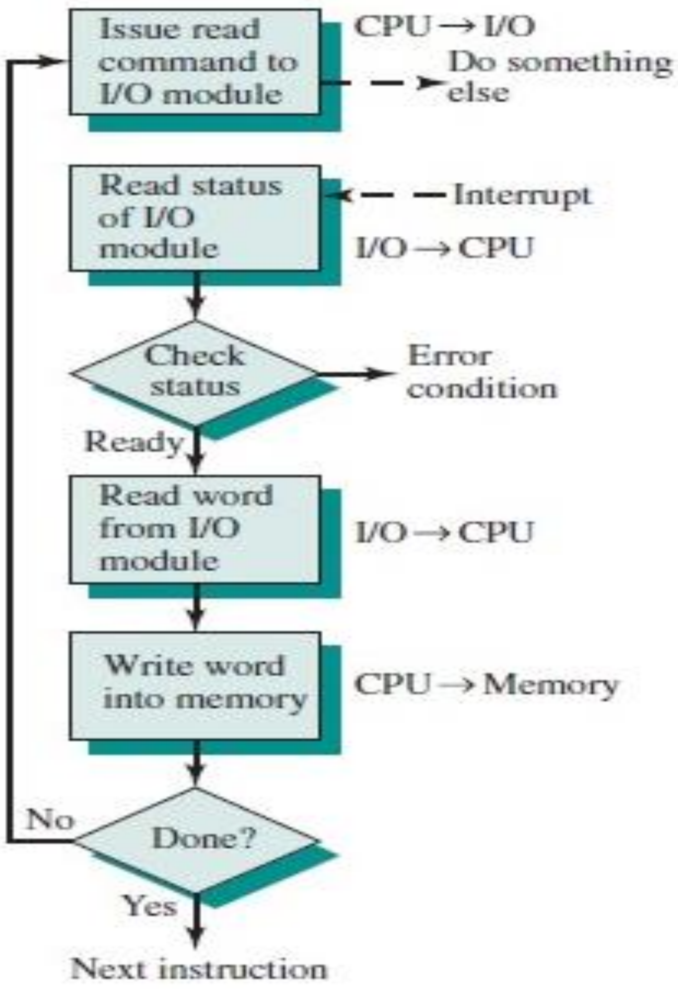
Interrupt driven I/O

- In Programmed I/O - CPU is kept busy unnecessarily – can be **avoided by using an interrupt** driven method for data transfer.
- By using interrupt facility and special commands to inform the interface to issue an interrupt request signal whenever data is available from any device (The interface keeps monitoring the device – Not the CPU)
- **CPU → $\boxed{\text{Interface → I/O → Interface}}$ → $\boxed{\text{CPU -----→ Memory}}$**
  (no CPU intervention) **(needs CPU intervention)**
- In the meantime the CPU can proceed for any other program execution.
- Whenever it is determined that the device is ready for data transfer **it initiates an interrupt request signal to the computer.**
- Upon detection of an external interrupt signal - CPU stops momentarily the task that it was already performing, branches to the service program to process the I/O transfer, and then return to the task it was originally performing.

Interrupt driven I/O - Process

- CPU setups and start I/O operation
- CPU goes off to do other things
- When I/O is done, CPU is interrupted
- CPU handles the interrupt
- CPU resumes interrupted operation





Interrupt driven I/O - Process

- The CPU issues the command to the I/O module
- The CPU then continues with what it was doing
- The I/O module, like before, issues the command to the I/O device and waits for the I/O to complete
- Upon completion of one byte input or output, the I/O module sends an interrupt signal to the CPU
- The CPU finishes what it was doing, then handles the interrupt
- This will involve moving the resulting datum to its proper location on input
- Once done with the interrupt, the CPU resumes execution of the program
  - This is much more efficient than Programmed I/O as the CPU is not waiting during the (time-consuming) I/O process

Direct Memory Access (DMA)

- Is a method that allows an input/output (I/O) device to send or receive data directly to or from the main memory, by-passing the CPU to speed up memory operations
- Why DMA?
  - Interrupt driven and programmed I/O require active CPU intervention
  - Transfer rate is limited
  - CPU is tied up
- DMA Function
  - Additional Module (hardware) on bus
  - DMA controller takes over from CPU for I/O transfers
- The process is managed by a chip known as a DMA controller (DMAC)

Direct Memory Access (DMA)

Role of CPU in transfer of Information

- Peripheral device -> CPU -> memory (With CPU - Programmed & Interrupted I/O)
  - CPU limits the speed of transfer
- Peripheral device -> memory (Without CPU - DMA)
  - Transfer speed increases
  - Peripheral device manage the memory bus directly.
  - DMA (Direct memory Access)
  - CPU is idle

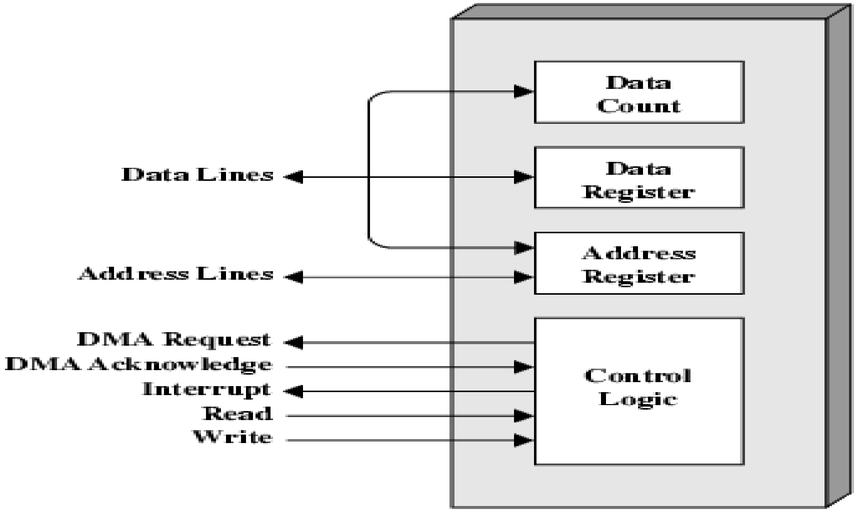
Interruped I/O: **CPU → Interface → I/O → Interface → CPU -----→ Memory**

DMA : **CPU → Interface → I/O → Interface → Memory**

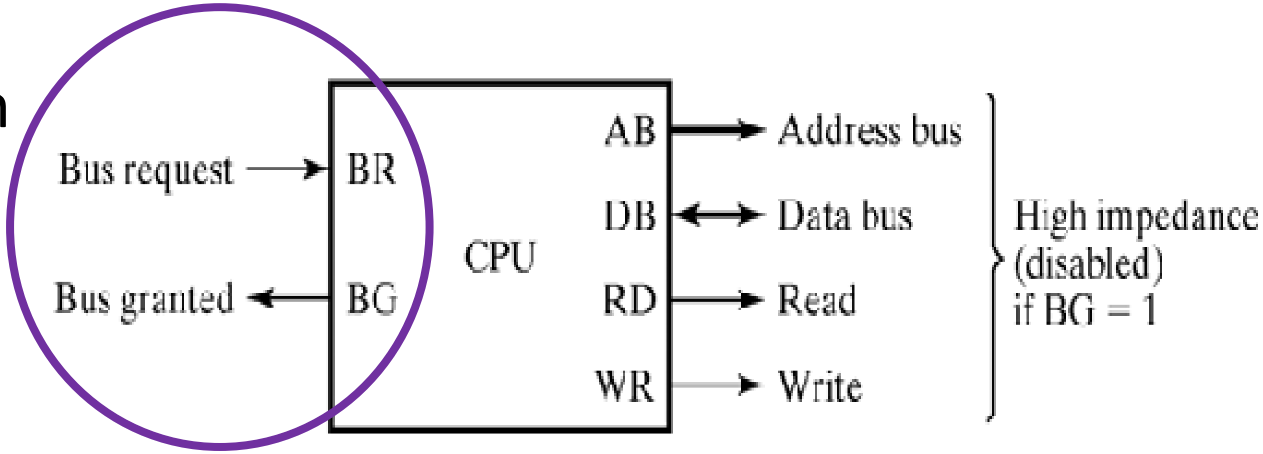
Direct Memory Access (DMA) - Module Design

CPU BUS Signals for DMA Transfer

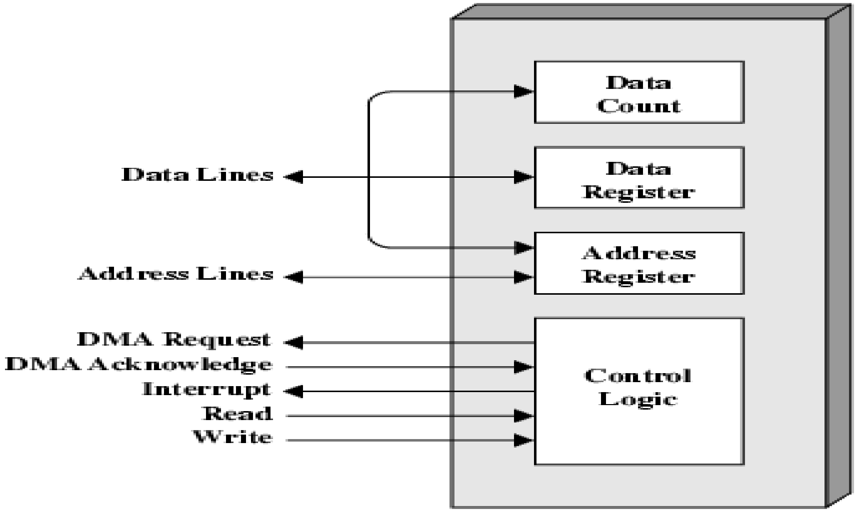
- **BR Signal**
- **BG Signal – issued by CPU**
  - **0** --- DMA communicates with CPU,
    1 ---- DMA communicates with Memory



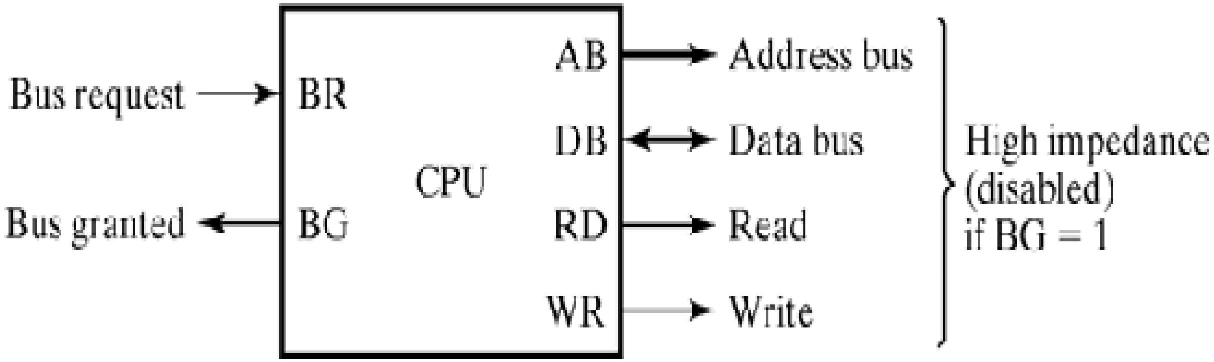
1. DMA Controller sends Bus Request to CPU
2. CPU stops execution of current instruction and places address bus, data bus, read and write lines into high impedance state.
3. CPU activates Bus Grant
4. After transfer, DMA disables BR (Bus Request)
5. CPU continues normal operation



Direct Memory Access (DMA) - Operation



1. CPU tells DMA controller:-
   - Read/Write
   - Device address
   - Starting address of memory block for data
   - Amount of data to be transferred
2. CPU carries on with other work
3. DMA controller deals with transfer
4. DMA controller sends interrupt when finished

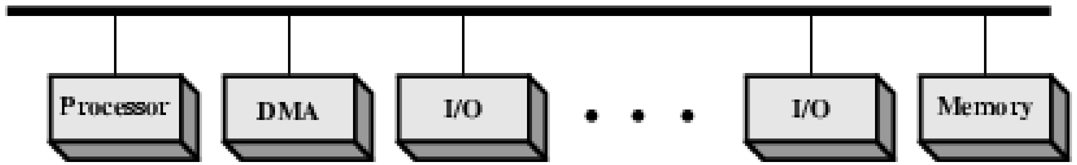


DMA - Different ways of DMA transfer

- **Burst transfer** – a block sequence consisting of a number of memory words is transferred in a continuous burst.
  - need for fast devices
- **Cycle stealing** – transfer one word at a time, after which it must return control of the buses to the CPU.
  - CPU delays 1 cycle to allow Direct Memory I/O transfer

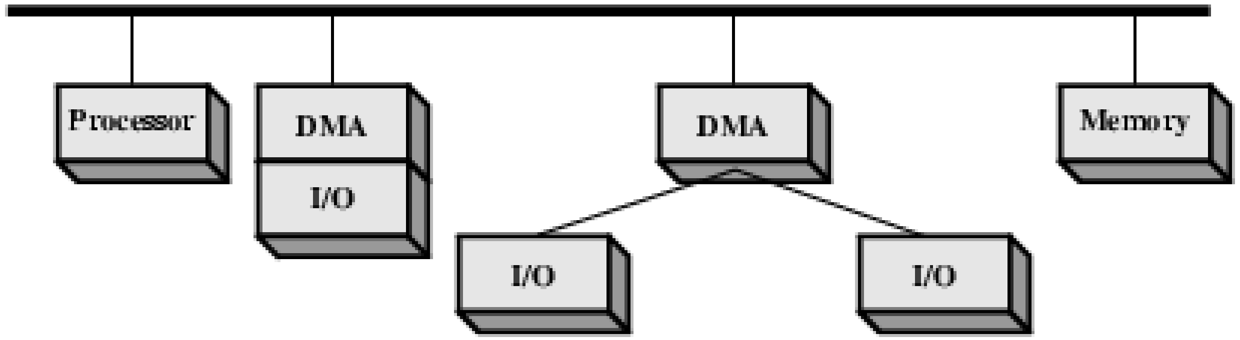
DMA - Configurations

Configuration 1: Single Bus, Detached DMA controller



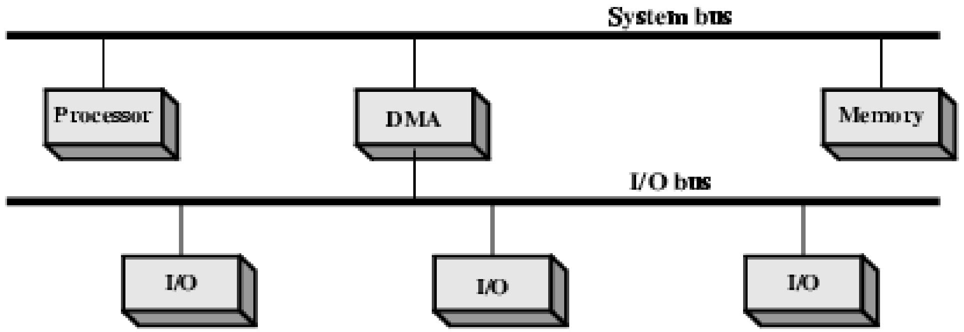
- Each transfer uses bus twice
    - I/O to DMA then DMA to memory
- CPU is suspended twice

Configuration 2: Single Bus, Integrated DMA controller

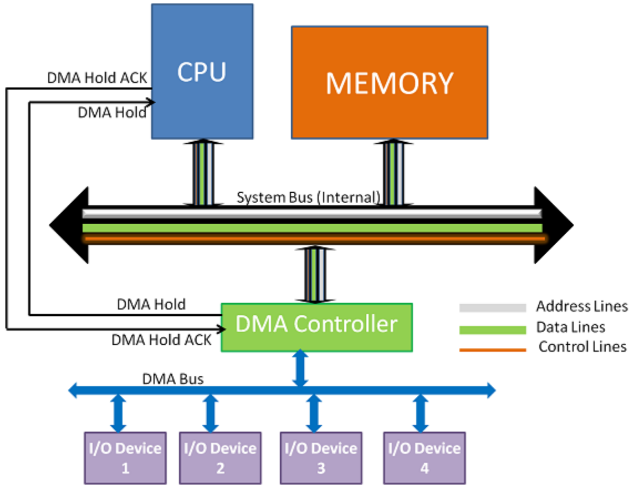


- Controller may support >1 device
- Each transfer uses bus once
    - DMA to memory
- CPU is suspended once

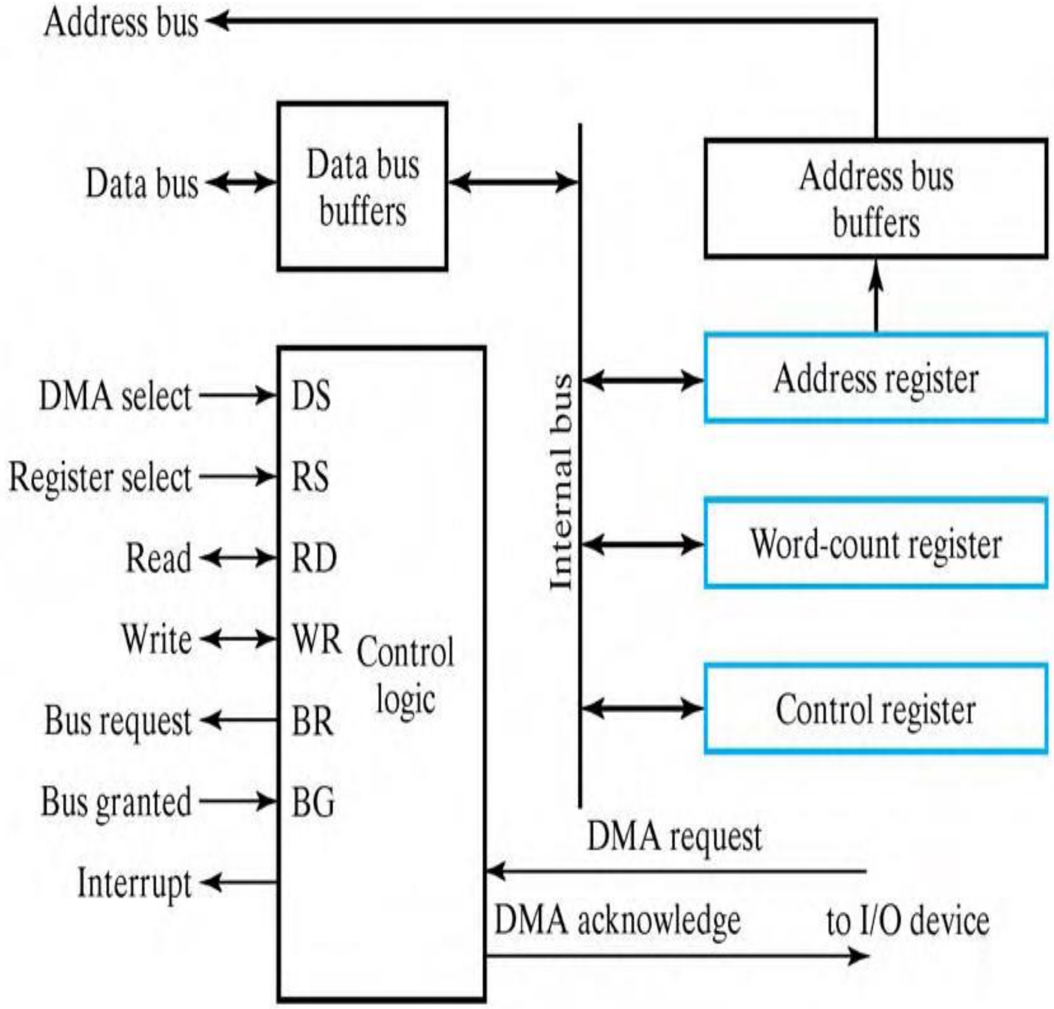
Configuration 3: Separate I/O Bus



- Bus supports all DMA enabled devices
- Each transfer uses bus once
    - DMA to memory
- CPU is suspended once

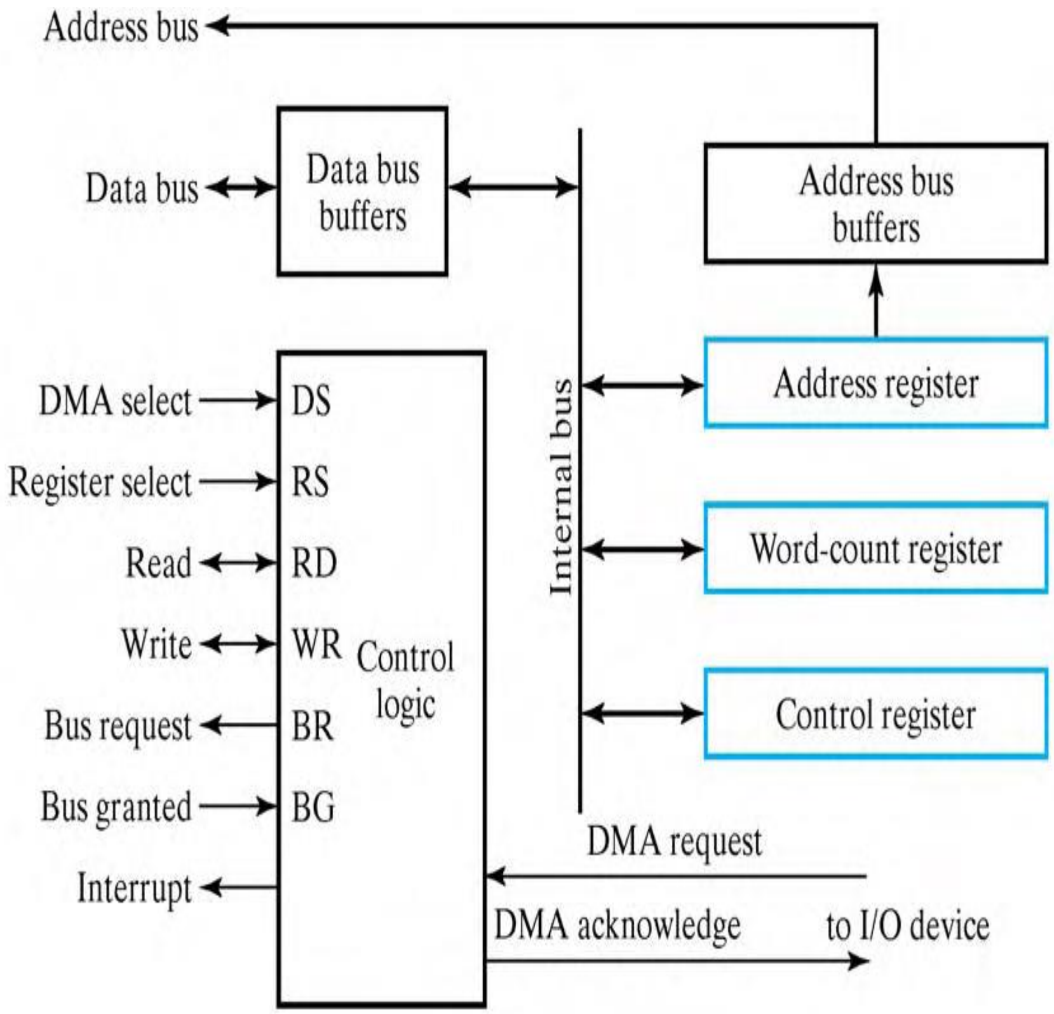


DMA Controller



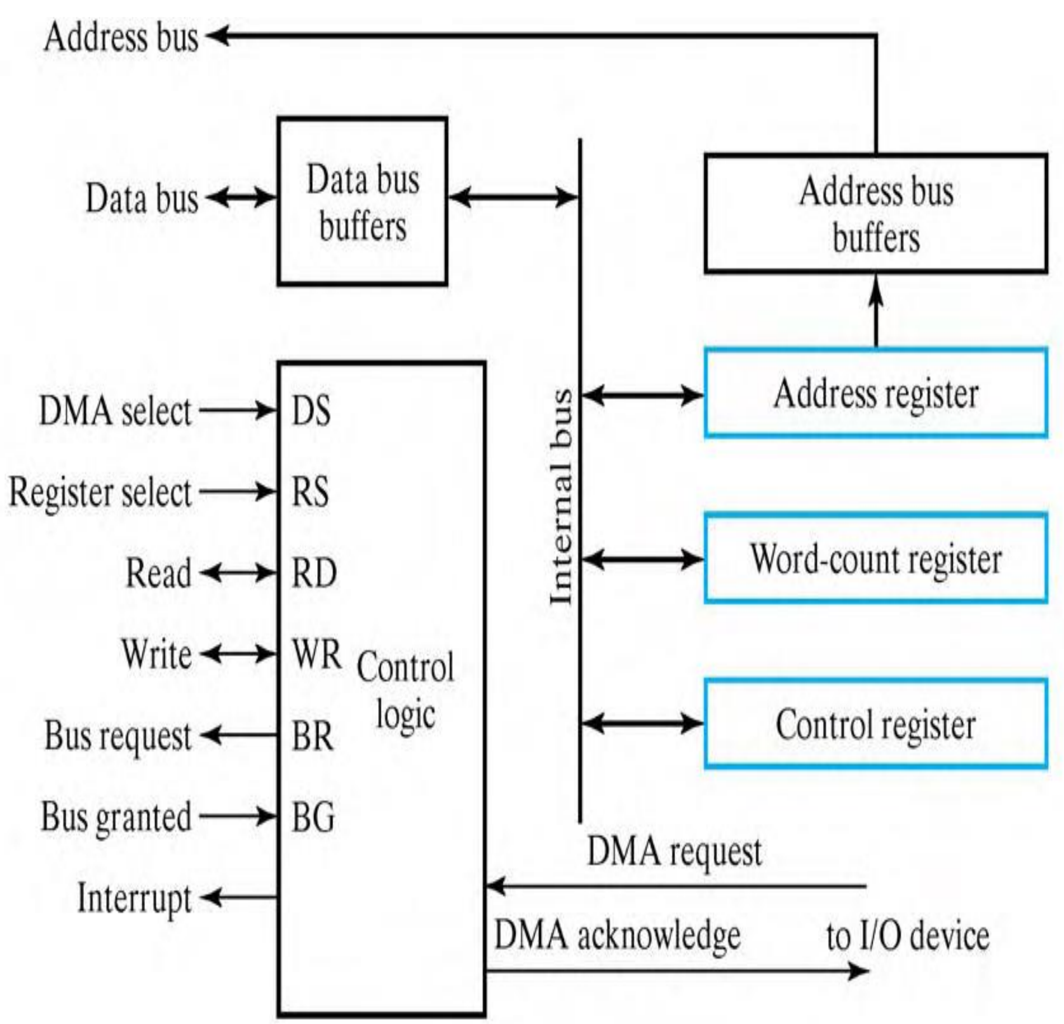
- Three registers
  - Address registers – to specify the desired location in memory
    - Incremented after each word transfer
  - Word count registers – numbers of words to be transferred
    - Decremented after each word transfer and tested for 0.
  - Control register – specifies mode of transfer

DMA Controller



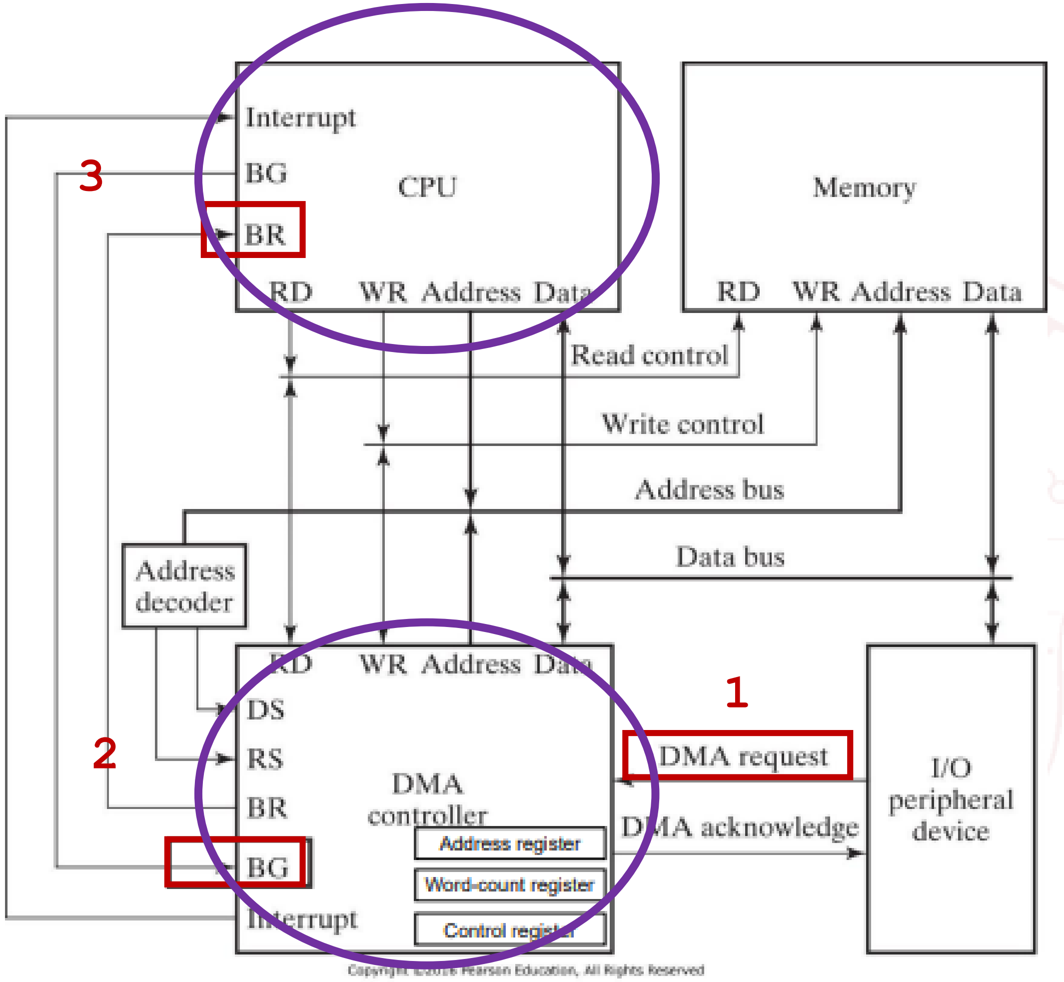
- Data bus and control lines => used to communicate with CPU
- Registers in DMA are selected by CPU through address bus by enabling DS and RS inputs.
- BG = 0 => CPU reads from or writes to the DMA registers
- BG =1 => DMA directly communicates with Memory by specifying address in address bus and activates the RD or WR control.
- DMA Request and acknowledge signals are used to communicate with peripheral devices

DMA Controller - Initialization



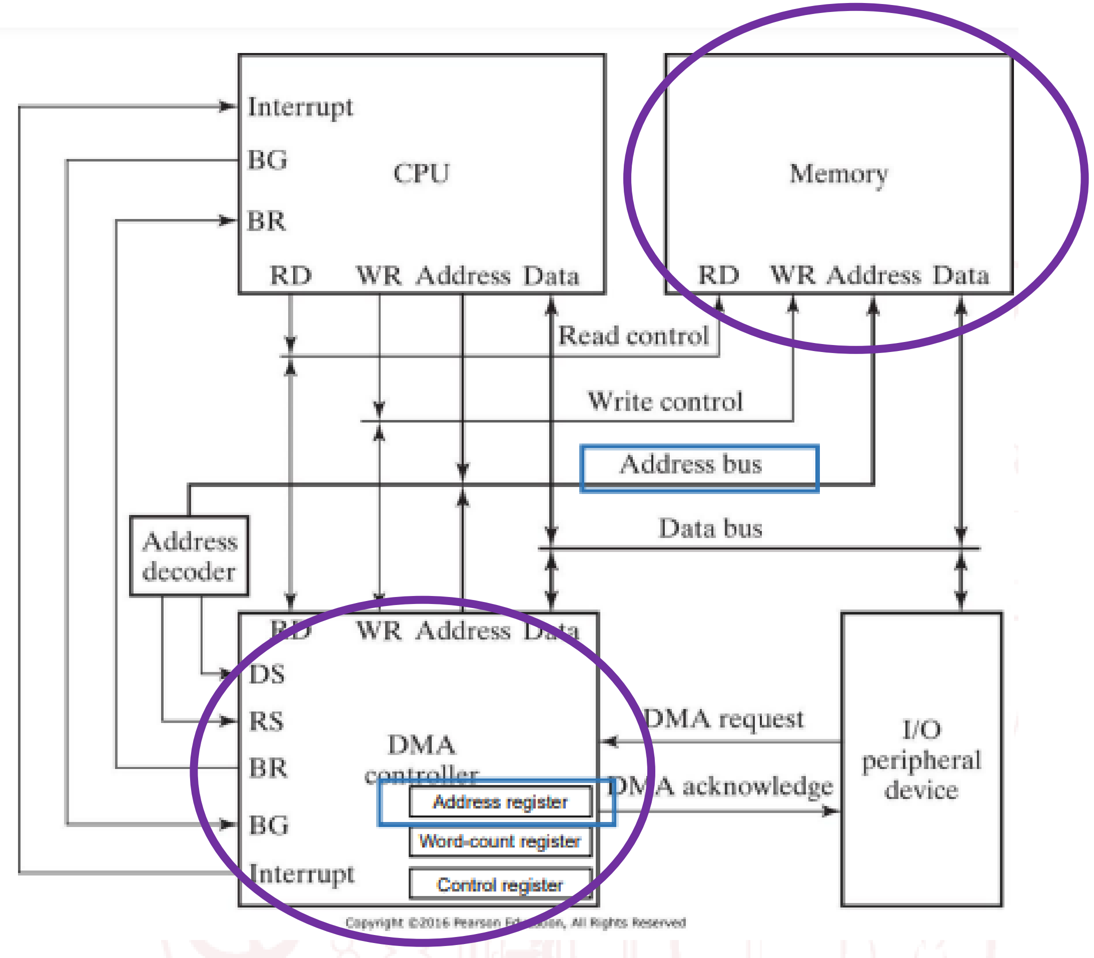
- CPU initializes DMA by sending the following information through data bus
  - Starting address of the memory block where data are available (for read) or where data are to be stored (for write)
  - The word count
  - Control – to read or write
  - A control to start the DMA transfer
- After initialization, CPU stops communicating with DMA unless it receives an interrupt signal or if it wants to checks how many words have been transferred.

DMA Transfer



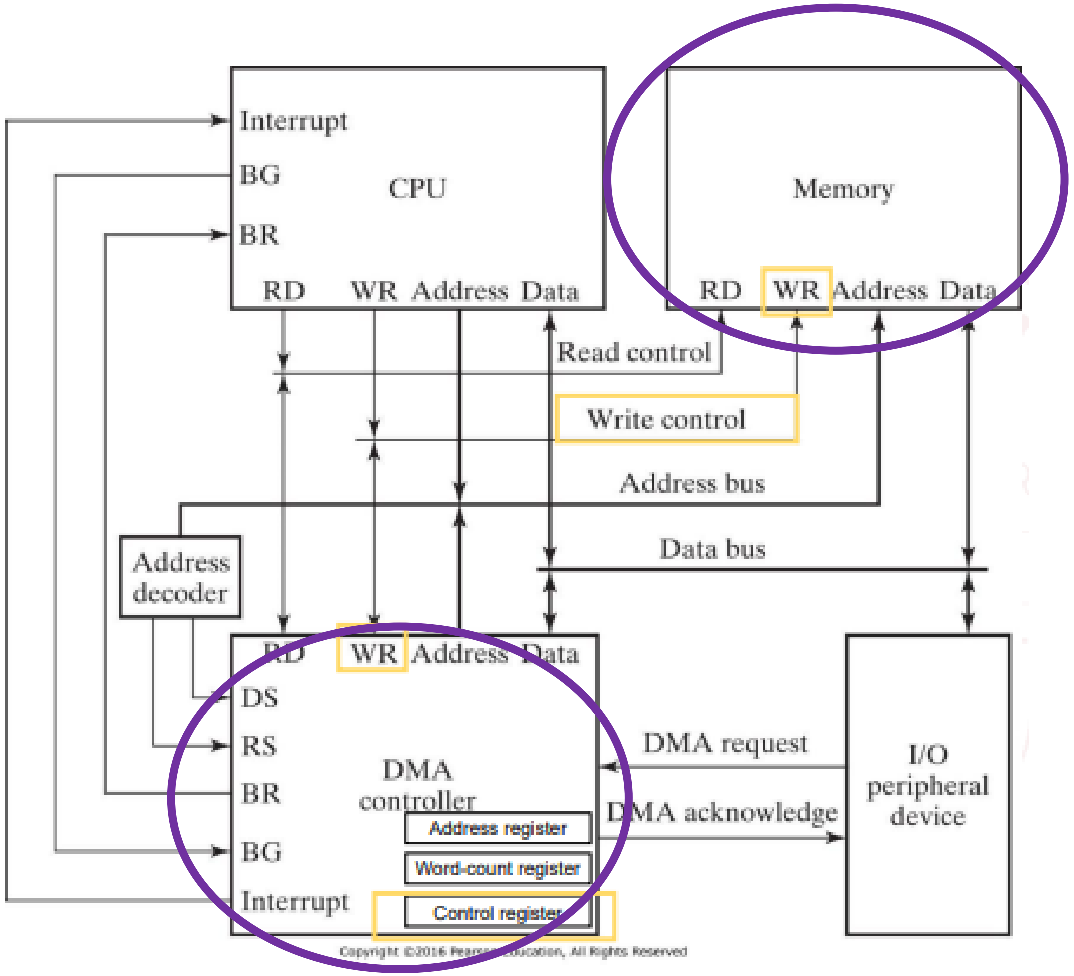
1. Peripheral device $\xrightarrow{\text{DMA Request}}$ DMA controller $\xrightarrow{\text{Bus Request}}$ CPU $\xrightarrow{\text{Bus grant}}$ DMA controller

DMA Transfer



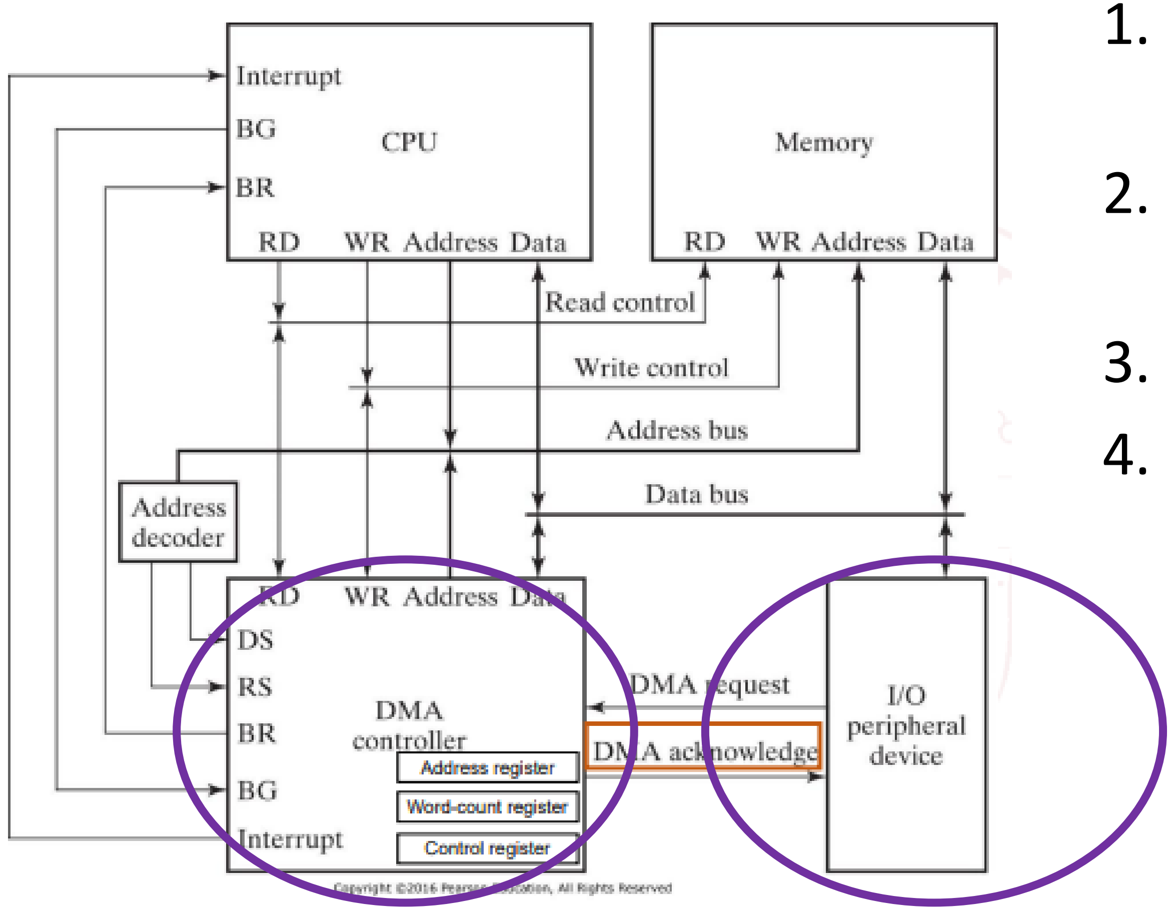
1. Peripheral device $\xrightarrow{\text{DMA Request}}$ DMA controller $\xrightarrow{\text{Bus Request}}$ CPU $\xrightarrow{\text{Bus grant}}$ DMA controller
2. DMA controller puts the current value of its address register onto address bus.

DMA Transfer



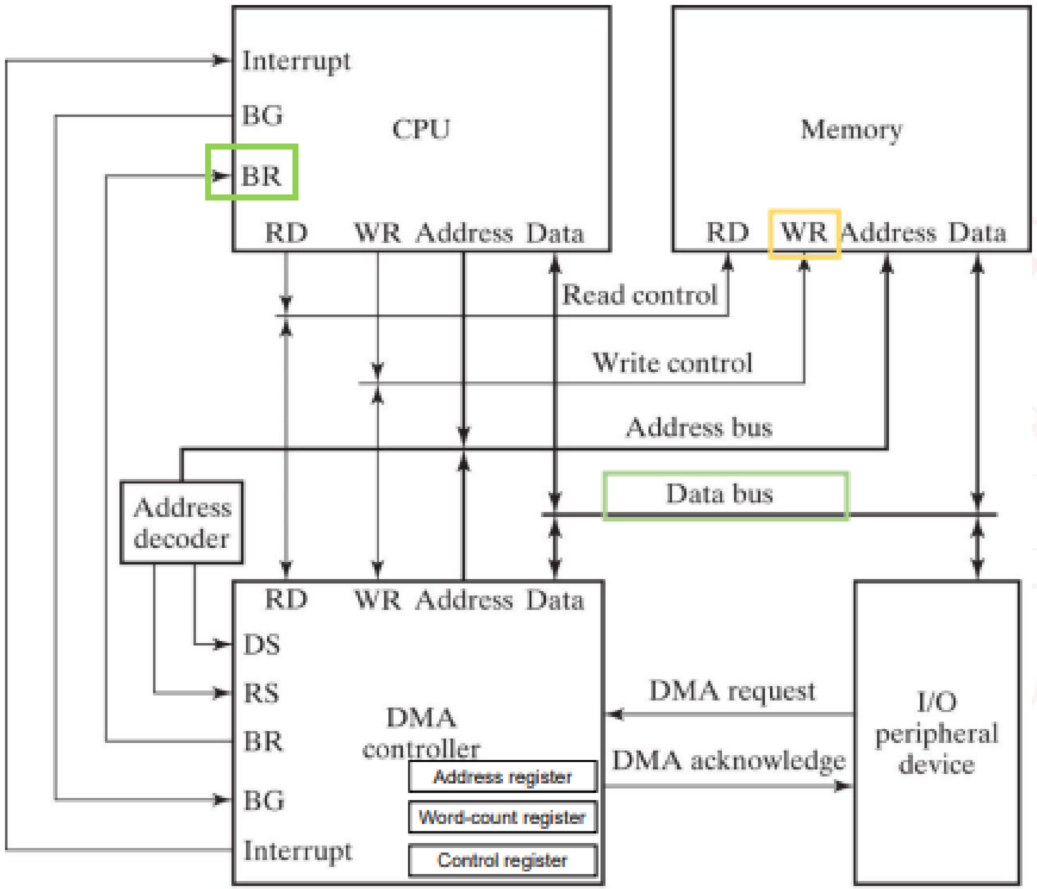
1. Peripheral device --DMA Request--> DMA controller --Bus Request--> CPU --Bus grant--> DMA controller
2. DMA controller puts the current value of its address register onto address bus.
3. DMAC activates RD or WR signal.

DMA Transfer



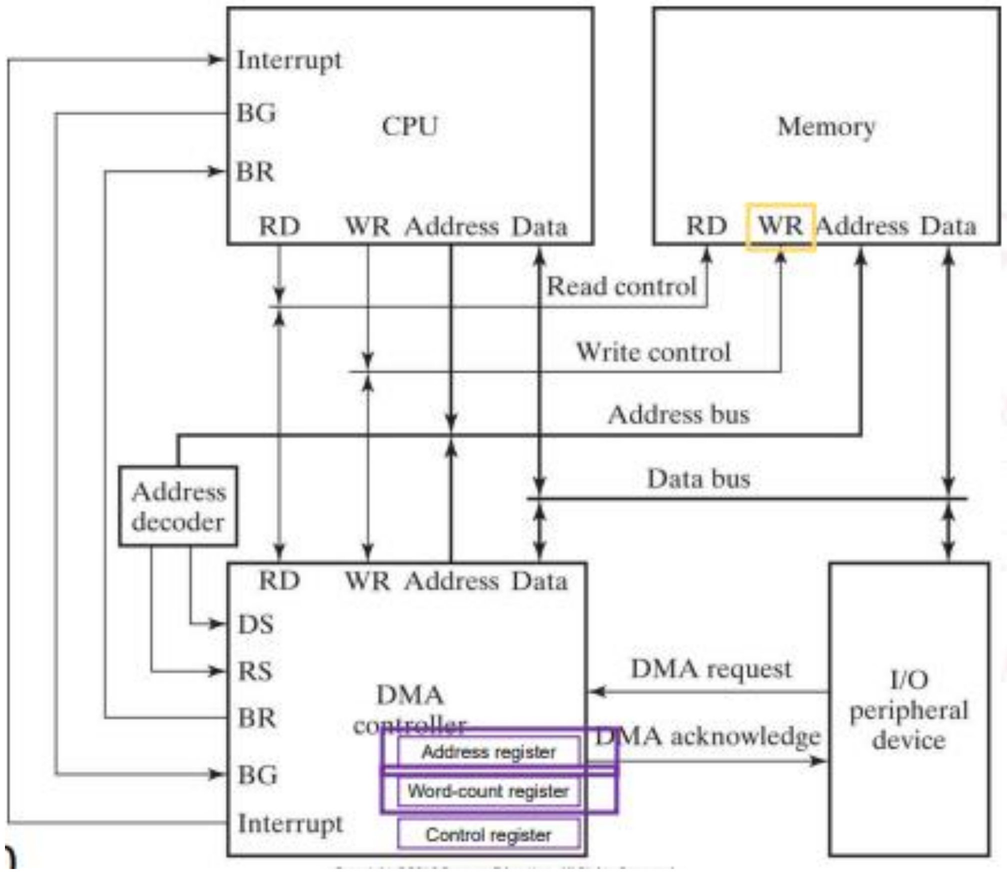
1. Peripheral device --DMA Request--> DMA controller --Bus Request--> CPU --Bus grant--> DMA controller
2. DMA controller puts the current value of its address register onto address bus.
3. DMAC activates RD or WR signal.
4. DMAC --DMA Acknowledge--> peripheral device

DMA Transfer



1. Peripheral device $\xrightarrow{\text{DMA Request}}$ DMA controller $\xrightarrow{\text{Bus Request}}$ CPU $\xrightarrow{\text{Bus grant}}$ DMA controller
2. DMA controller puts the current value of its address register onto address bus.
3. DMAC activates RD or WR signal.
4. DMAC $\xrightarrow{\text{DMA Acknowledge}}$ peripheral device
5. RD and WR are bidirectional
   1. BR = 0 => CPU communicates with the internal DMA registers
   2. BR = 1 => RD and WR are output lines from DMA controller to the RAM to specify read or write operation for data

DMA Transfer



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DMA - Application

- Fast transfer of information between magnetic disk and memory
- Updating the display in an interactive terminal

DMA - Problems

1 Byte	=	8 bits		
1 KB	=	1024 B(ytes)	=	2^{10} B
1 MB	=	1024 KB	=	2^{20} B
1 GB	=	1024 MB	=	2^{30} B

Problem 1:

A hard disk with a transfer rate of 20 Mbytes/second is constantly transferring data to memory using DMA. The processor runs at 300MHz, and takes 300 and 900 clock cycles to initiate and complete DMA transfer respectively. If the size of the transfer is 20 Kbytes, what is the percentage of processor time consumed for the transfer operation?

Solution:

$$Total\ Time = \frac{Total\ Data}{Total\ Transfer\ Rate}$$

$$Processor\ Time = \frac{Cycles\ for\ DMA\ transfer}{Processor\ Speed}$$

$$\%\ Processor\ Time = \frac{Processor\ Time}{Total\ Time} \times 100$$

Step 1: Find Total Transfer Time

Transfer rate=20 MB per second = 20×2^{20} bytes/sec

Data=20 KB=20×2^{10} bytes

Time=$(20 \times 2^{10})/(20 \times 2^{20})= 1 \times 2^{-10} = 1 \times 10^{-3}$ =1 ms

Step 2 :Find Processor Time

1 MHz = 10^6 cycles/sec

Processor speed= 300 MHz = 300×10^6 Cycles/sec

Cycles required by CPU for DMA Transfer =300+900 =1200 cycles

Time=1200 cycles/(300×10^6) cycles/sec = 4×10^{-6} sec = .004 ms

Step 3 : Find % of Processor Time

% Processor Time = (.004/1) × 100 = 0.4%

Term	Normal Usage	Usage as Power of 2
K (Kilo)	10^3	2^{10} = 1,024
M (Mega)	10^6	2^{20} = 1,048,576
G (Giga)	10^9	2^{30} = 1,073,741,824
T (Tera)	10^{12}	2^{40} = 1,099,511,627,776

1 Byte	= 8 bits		
1 KB	= 1024 B(ytes)	=	2^{10} B
1 MB	= 1024 KB	=	2^{20} B
1 GB	= 1024 MB	=	2^{30} B

DMA - Problems

Problem 2:

A hard disk with a transfer rate of 10 Mbytes/second is constantly transferring data to memory using DMA. The processor runs at 600MHz, and takes 300 and 900 clock cycles to initiate and complete DMA transfer respectively. If the size of the transfer is 20 Kbytes, what is the percentage of processor time consumed for the transfer operation?

$$Total\ Time = \frac{Total\ Data}{Total\ Transfer\ Rate}$$

$$Processor\ Time = \frac{Cycles\ for\ DMA\ transfer}{Processor\ Speed}$$

$$\%\ Processor\ Time = \frac{Processor\ Time}{Total\ Time} \times 100$$

Solution:

Step 1: Find Total Transfer Time

Transfer rate=10 MB per second = 10×2^{20} bytes/sec

Data=20 KB=20×2^{10} bytes

Time=$(20 \times 2^{10})/(10 \times 2^{20})= 2 \times 2^{-10} = 2 \times 10^{-3} = 2$ ms

Step 2 :Find Processor Time

1 MHz = 10^6 cycles/sec

Processor speed= 600 MHz = 600×10^6 Cycles/sec

Cycles required by CPU for DMA Transfer =300+900 =1200 cycles

Time=1200 cycles/(600×10^6) cycles/sec = 2×10^{-6} sec = .002 ms

Step 3 : Find % of Processor Time

% Processor Time = (.002/2) × 100 = 0.1%

Term	Normal Usage	Usage as Power of 2
K (Kilo)	10^3	$2^{10} = 1,024$
M (Mega)	10^6	$2^{20} = 1,048,576$
G (Giga)	10^9	$2^{30} = 1,073,741,824$
T (Tera)	10^{12}	$2^{40} = 1,099,511,627,776$